AYE

• INDUSTRIAL ROBOTICS • LESSON 06

# Fanuc Robot Systems

## R-30iB Plus • TP Programming • J/L/C Motion • Registers

Robot Models | Controller | Teach Pendant | Comparison with KUKA & ABB

### What You Will Learn

- Fanuc robot series: LR Mate, M-10iA, M-20iB, R-2000iC, M-2000iA — payload/reach/application
- R-30iB Plus controller: components, teach pendant (iPendant), operation modes
- TP programming: J/L/C motion instructions, speed, zone (FINE/CNT), position registers
- Fanuc vs KUKA vs ABB — direct comparison of programming language and robot control concepts

### FANUC TP Motion Example

```
3:  J  PR[1:HOME]  30% FINE  ;
4:  L  P[1:APPROACH]  500mm/sec  FINE  ;
```

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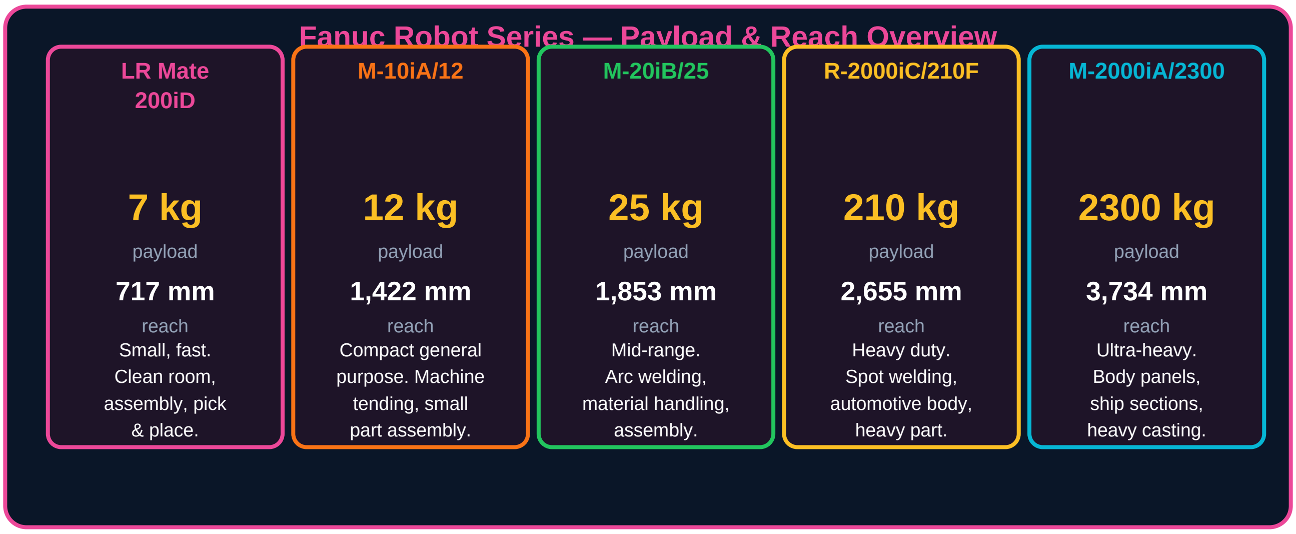
# ROBOT-006: Fanuc Robot Systems

Fanuc Corporation, headquartered in Oshino, Japan, is the world's largest manufacturer of CNC systems and industrial robots by market share (approximately 30% globally in 2026). Fanuc robots are recognised by their distinctive yellow colour and are the dominant platform in automotive manufacturing, electronics, food and beverage, and metal fabrication worldwide. Their R-30iB Plus controller and TP (Teach Pendant) programming language are industry standards that every automation engineer is likely to encounter on the job.

## Learning Objectives

- Identify the main Fanuc robot series and select the appropriate model for a given application
- Describe the R-30iB Plus controller architecture and navigate the iPendant teach pendant
- Write TP programs using J, L, and C motion instructions with speed, zone, and tool settings
- Use position registers (PR[]) and numeric registers (R[]) for flexible robot programming
- Compare Fanuc TP programming concepts with KUKA KRL and ABB RAPID

## 1. Fanuc Robot Series




| Series | Payload Range | Reach Range | Key Characteristic | Typical Application |
|---|---|---|---|---|
| LR Mate 200iD | 7 – 14 kg | 717 – 911 mm | Smallest footprint. IP67 food-grade version available. Highest speed in class. | Assembly, testing, dispensing, clean room, small part handling |
| M-10iA / M-10iD | 10 – 12 kg | 1,097 – 1,422 mm | Slim arm for confined spaces. Wall/ceiling mounting option. | Machine tending, arc welding, material removal |

| | | | | |
|---|---|---|---|---|
| M-20iB | 20 – 35 kg | 1,813 – 1,850 mm | Most versatile mid-range. Highest rigidity in class for dynamic applications. | Arc welding, assembly, material handling, palletising |
| R-2000iC | 100 – 270 kg | 2,438 – 2,655 mm | Spot welding workhorse. Dominant in automotive body-in-white. | Spot welding, press loading, automotive assembly |
| M-710iC | 50 – 70 kg | 2,050 – 2,354 mm | Long reach with high payload ratio. Hollow wrist for through-arm dressing. | Arc welding, large part handling, surface treatment |
| M-2000iA | 900 – 2,300 kg | 3,734 mm | World's strongest robot. Lifts car body shells, press dies, ship sections. | Automotive stamping, heavy foundry, ship hull assembly |

## 2. R-30iB Plus Controller

The R-30iB Plus (A05B-2600-H002) is Fanuc's current generation controller, introduced in 2014 and continuously updated. It is available in three form factors: standard cabinet (mate, B, and B Plus), compact cabinet (A cabinet for smaller robots), and panel-mount (for integration into customer cabinets). All share the same software — iRVision, force sensing, DCS safety, and EtherNet/IP connectivity are available across the range.

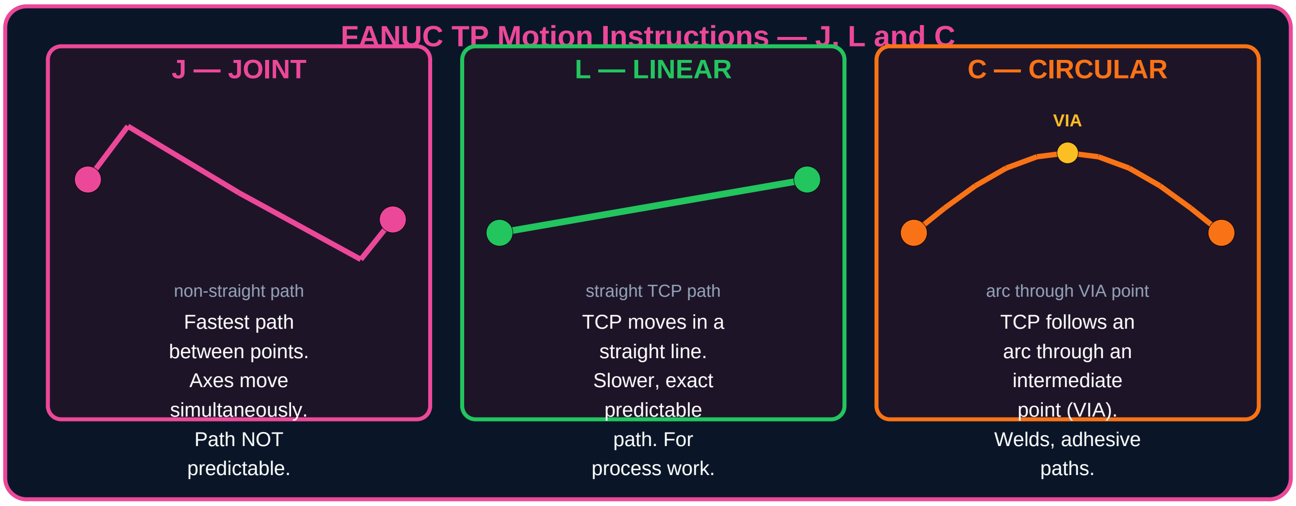
| Controller Component | Function | Technical Notes |
|---|---|---|
| Main CPU board | Runs the robot motion planning, program execution, and safety monitoring | Dual-core processor. Cycle time: 1ms servo, 8ms ladder. Real-time OS (FROS). |
| Servo amplifier module | Converts DC bus to 3-phase AC for each servo motor (1 amplifier per axis) | 6-axis standard. Up to 3 additional external axes with expansion modules. |
| Power supply (PSU) | Rectifies 3-phase 200–480V AC to DC bus. Powers all servo amplifiers. | Integrated regenerative braking — braking energy returned to grid, not wasted as heat. |
| DCS (Dual Check Safety) | Dual-channel safety monitoring: speed limits, work envelope, tool orientation limits | SIL 2 / PLd certified. Replaces hardwired safety devices in many applications. |
| Emergency stop circuit | Category 3 / PLe safety circuit. Auto-detects broken teach pendant cable. | Internal and external E-stop inputs. E-stop on teach pendant AND external button required. |
| I/O unit (CRMA15/16) | Digital inputs and outputs (DI/DO) for PLC and peripheral device communication | Default: 28 DI + 20 DO. Expandable to 512 DI/DO with Process I/O board. |
| EtherNet/IP adapter | Industrial Ethernet for PLC communication (Rockwell, Siemens, Beckhoff, etc.) | 100Mbps. Supports tag-based communication. Enables remote program start and status monitoring. |

## iPendant (Teach Pendant) — Key Controls

The Fanuc iPendant is a colour touch-screen pendant with a 4-axis jog key (4 arrows + SHIFT key for axis selection), a 3-position deadman switch, and an emergency stop button. The screen shows the current program, position data, and system status. Unlike KUKA's hand-guided teach, Fanuc's standard jogging uses the individual axis keys.

| Control / Key | Function |
|---|---|
| 3-position deadman (safety switch) | Middle position = robot motion enabled in T1/T2 mode. Release or squeeze fully = immediate stop. Must be held for any robot movement in teach modes. |
| SHIFT key | Activates robot motion in T1/T2 when combined with jog keys. Hold SHIFT + jog key to move robot. Release SHIFT immediately stops motion. |
| COORD key | Selects coordinate system for jogging: JOINT (each axis independently), WORLD (XYZ global), TOOL (relative to tool face), USER (relative to user frame). |
| F1–F5 function keys | Context-sensitive menu options shown above each key on screen. Change with NEXT key to access more functions. |
| STEP key | Toggles between single-step and continuous execution mode for test running programs. |
| SELECT / EDIT / DATA | Navigate between program list (SELECT), open program editor (EDIT), and system data screens (DATA). |
| BWD / FWD | Step program backward (BWD) or forward (FWD) one line at a time when testing. Test speed limited to OVERRIDE setting. |

## 3. TP Motion Instructions




### TP Motion Instruction Syntax

**Line number : Motion_type Position Speed Zone [options] ;**

| Instruction | Syntax | Speed Unit | When to Use |
|---|---|---|---|
| J (Joint) | J P[1] 50% FINE | % of max joint speed | Moving between non-critical positions. Fastest, joints move simultaneously. Path not controlled — avoid obstacles by choosing intermediate points. |
| J with PR | J PR[1:HOME] 30% FINE | % of max joint speed | Moving to a named position register (flexible — can change position without editing program). Use for home, park, and changeable pick points. |
| L (Linear) | L P[2] 500mm/sec FINE | mm/sec or cm/min | Any move where TCP must travel in a straight line: welding, dispensing, insertion, pick and place with orientation change. |
| L with CNT | L P[3] 800mm/sec CNT100 | mm/sec | Blended move — robot approaches P[3] and smoothly transitions to the next move without stopping. CNT100 = maximum blending radius. |
| C (Circular) | C P[via] P[end] 300mm/sec FINE | mm/sec | Arc path through a via point (P[via]) to end point (P[end]). Used for circular welds, circular dispensing paths, curved surface following. |

### Zone Options — FINE vs CNT

| Zone | Behaviour | Use When |
|---|---|---|
| FINE | Robot decelerates to exact stop at the taught point. Zero path deviation. Slowest for multi-point paths. | Pick, place, welding start/stop, any position requiring precision. Safety-critical positions like tool change. |

| | | |
|---|---|---|
| CNT 0 | Equivalent to FINE — also stops at the point. | Same as FINE — use FINE for clarity. |
| CNT 1–100 | Robot blends the path through the point. CNT100 = largest blend, CNT1 = smallest blend. Never fully reaches the point — path passes near it. | Intermediate approach or retract points where exact position is not needed. Reduces cycle time significantly. |

## 4. Registers and Program Structure

**FANUC TP Program Structure & Register System**

**TP PROGRAM STRUCTURE**

```
/PROG  PICK_PART
/ATTR
OWNER = MNEDITOR
COMMENT = "Pick cycle"
/MN
 1:  UTOOL_NUM = 1   ;
 2:  UFRAME_NUM = 0  ;
 3: J PR[1:HOME] 30% FINE ;
 4: L P[1:APPROACH] 500mm/sec FINE ;
 5: L P[2:PICK]     100mm/sec FINE ;
 6:  DO[1]=ON  ;
 7:  WAIT  0.50sec ;
 8: L P[1:APPROACH] 500mm/sec FINE ;
 9: J PR[1:HOME] 30% FINE ;
/END
```

**REGISTER TYPES**

- **R[n]** — Numeric register; Integer or REAL value
- **PR[n]** — Position register; X,Y,Z,W,P,R coordinates
- **DI[n]/DO[n]** — Digital input/output; Control external devices
- **AI[n]/AO[n]** — Analogue I/O; 4-20mA, 0-10V signals

## Using Position Registers for Flexible Programming

Position registers (PR[n]) are the key to flexible, intelligent robot programs. Instead of hard-coding every pick position in the program, the program reads the position from a register that can be calculated, received from a PLC, or modified without changing the program structure.

| Operation | TP Code | Description |
|---|---|---|
| Set R register | R[1]=5 | Assign numeric value 5 to register R[1] |
| Arithmetic | R[2]=R[1]+R[3] | Add R[1] and R[3], store result in R[2] |
| Copy position to PR | PR[2]=LPOS | Copy current Cartesian position to PR[2] |
| Offset position | PR[3]=PR[2] PR[3,3]=PR[3,3]+50 | Offset Z-axis by 50mm (component 3 = Z) |
| Move to PR | L PR[5] 500mm/sec FINE | Move linearly to position in PR[5] |
| Conditional | IF R[1]=1, JMP LBL[1] | Jump to label 1 if R[1] equals 1 |
| I/O read | IF DI[1]=ON, JMP LBL[2] | Jump if digital input 1 is ON |
| I/O write | DO[1]=ON | Turn digital output 1 ON (e.g. gripper close) |
| Wait for input | WAIT DI[1]=ON | Pause program until DI[1] goes ON (PLC handshake) |
| Call macro | CALL GRIP_CLOSE | Call a sub-program (macro) named GRIP_CLOSE |

## 5. Fanuc vs KUKA vs ABB — Direct Comparison

**Three Major Robot Platforms — Key Differences**

| FANUC | KUKA | ABB |
|---|---|---|
| R-30iB Plus | KRC4 | IRC5 / OmniCore |
| **TP (Teach Pendant) Language** | **KRL (KUKA Robot Language)** | **RAPID Language** |
| J/L/C motion<br>PR[] position reg<br>DI/DO I/O<br>MACRO programs<br>ROSET/LPOS | PTP / LIN / CIRC<br>KRC variables<br>E6POS data<br>SRC + DAT files<br>INTERRUPT | MoveJ/MoveL/<br>MoveC motion<br>robtarget pos.<br>Tooldata<br>Procedure/Func |

**All three use IEC 61131-3 logic for I/O. Programming concepts transfer between brands — only the syntax changes.**

Market share 2026: Fanuc ~30% | KUKA ~18% | ABB ~16% | Others ~36%

## Fanuc vs KUKA vs ABB — Programming Syntax Comparison

| Concept | Fanuc TP | KUKA KRL | ABB RAPID |
|---|---|---|---|
| Move joint | J P[1] 50% FINE | PTP P1 Vel=50% PDAT1 | MoveJ p1, v500, fine, tool1 |
| Move linear | L P[2] 500mm/sec FINE | LIN P2 Vel=500 CPDAT1 | MoveL p2, v500, fine, tool1 |
| Move circular | C P[via] P[end] 300mm/sec FINE | CIRC P_aux P_end Vel=300 CPDAT | MoveC via, end, v300, fine, tool1 |
| Digital output ON | DO[1]=ON | $OUT[1]=TRUE | Set do1; |
| Wait for input | WAIT DI[1]=ON | WAIT FOR $IN[1]==TRUE | WaitDI di1, 1; |
| Numeric variable | R[1]=100 | DECL INT speed=100 | VAR num speed := 100; |
| Position variable | PR[1] (register) | DECL E6POS myPos | VAR robtarget myPos; |
| Call subprogram | CALL SUBPROG | subprog() | PROC SubProg() ... ENDPROC |
| Loop | LBL[1] ... JMP LBL[1] | LOOP ... ENDLOOP | WHILE TRUE DO ... ENDWHILE |

## 6. Practice Exercises

| # | Exercise | Task |
|---|---|---|
| 1 | Model selection | A food packaging line requires a robot to pick blister packs (380g each) from a conveyor and place them in a box. Reach needed: 900mm. Cycle time: 60 picks/min. Environment: food-grade. Which Fanuc series would you select? Justify your choice based on payload, reach, and food-grade requirements. |
| 2 | TP program writing | Write a Fanuc TP program for a pick-and-place cycle: (a) Move joint to PR[1:HOME] at 30% speed FINE. (b) Move linear to P[1:APPROACH] at 800mm/sec CNT50. (c) Move linear to P[2:PICK] at 200mm/sec FINE. (d) Turn DO[1]=ON (gripper close). Wait 0.5 seconds. (e) Move linear to P[1:APPROACH] at 500mm/sec FINE. (f) Move joint to PR[1:HOME] at 30% speed FINE. |
| 3 | Zone selection | For each position in the following pick-and-place cycle, state whether to use FINE or CNT and why: (a) Home position. (b) Approach point 100mm above pick. (c) Pick point. (d) Retract point 100mm above pick. (e) Place approach 50mm above target. (f) Place point. |

| | | |
|---|---|---|
| 4 | Register programming | A palletising program must place parts at X=500mm, Y=300mm offset from PR[1:PALLET_ORIGIN]. Write the TP code to: (a) Copy PR[1] to PR[10]. (b) Add 500mm to X component (PR[10,1]). (c) Add 300mm to Y component (PR[10,2]). (d) Move linearly to PR[10] at 300mm/sec FINE. |
| 5 | Platform comparison | A client's factory has KUKA KRC4 robots. They are installing a new Fanuc R-2000iC for a different cell. List: (a) 3 programming concepts that transfer directly from KUKA KRL to Fanuc TP. (b) 3 major syntax differences an engineer must learn when switching platforms. (c) Which platform would you recommend for a new engineer learning robot programming for the first time, and why? |

## Lesson Summary

| Key Concept | Summary Rule |
|---|---|
| Fanuc robot selection | LR Mate: small/fast/food. M-10/20: general mid-range. R-2000: spot welding/heavy. M-2000: ultra-heavy. All use same R-30iB Plus controller and TP language. |
| R-30iB Plus controller | Dual-channel safety (DCS). 6-axis standard, expandable to 9. EtherNet/IP for PLC communication. iPendant with touch screen. |
| J motion | Joint interpolation — fastest, not predictable path. Use for non-critical moves between positions. Always check for collisions with intermediate points. |
| L motion | Linear TCP path — controlled, predictable. Required for welding, dispensing, insertion. Speed in mm/sec. Slower than J but essential for process work. |
| C motion | Circular arc through a VIA point to end point. Two position arguments: C P[via] P[end]. Used for circular welds and curved paths. |
| FINE vs CNT | FINE = exact stop at point (slower, precise). CNT 1–100 = blended path through point (faster, not exact). Use FINE for pick/place, CNT for approach/retract. |
| Position registers PR[] | Flexible positions that can be modified by program logic. Essential for palletising, array picking, and offset calculations. |
| DO[]/DI[] I/O | DO[n]=ON/OFF to control outputs (gripper, conveyor). WAIT DI[n]=ON to synchronise with PLC handshake signals. |

## What Is Next — ROBOT-007

**ROBOT-007: End Effectors and Grippers** — Types of end effectors: pneumatic grippers, electric servo grippers, vacuum cups, magnetic grippers, and custom tooling. Gripper selection based on part weight, shape, material, and cycle time. Force and torque considerations. Tool Centre Point (TCP) calibration procedure. Tool data setup in KUKA, Fanuc, and ABB systems.

| Resource | Details |
|---|---|

| | |
|---|---|
| Robotics PDF Series | ROBOT-001 through ROBOT-030 — ayetechub.com/pdfs.html |
| Fanuc iRVision & Sim | ROBOGUIDE simulation — free trial at fanuc.com/products/roboguide |
| Community | Telegram: t.me/ayetechub |